



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 5 (Summary and Tutorial)

Topic: Complex Numbers

Summary for Complex Numbers

Definition of Imaginary Numbers i

$i = \sqrt{-1}$

Note: $i^2 = -1$ $i^3 = (i^2)(i) = -i$ $i^4 = (i^2)^2 = 1$

Representation of a complex number		Conjugate
Cartesian Form	$z = x + iy$	$z^* = x - iy$
Argand diagram	$z = x + iy$ can be represented as • point P with coordinates (x, y) • $z = r = \sqrt{x^2 + y^2}$ • $\arg(z) = \theta$, where $-\pi < \theta \leq \pi$	z^* can be represented as • point P' with coordinates $(x, -y)$, which is the reflection of P in the real axis • $z^* = r = \sqrt{x^2 + y^2}$ • $\arg(z^*) = -\theta$,
	Eg: for the case where $x > 0$ and $y > 0$:	

Equality of Complex Numbers

2 complex numbers are equal if and only if

$$z_1 = z_2 \Leftrightarrow x_1 = x_2 \text{ and } y_1 = y_2 \quad (\text{equate real and imaginary parts})$$

where $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, $x_1, y_1, x_2, y_2 \in \mathbb{R}$.

Useful properties involving Conjugates

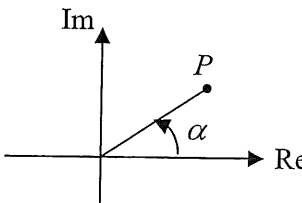
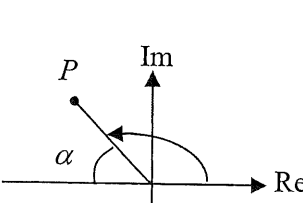
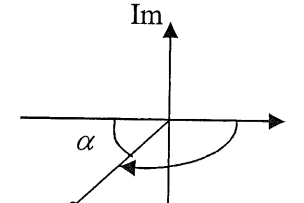
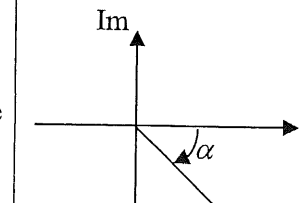
Consider a complex number $z = x + iy$, $x, y \in \mathbb{R}$, then

(a) $(z^*)^* = z$	(e) $z = z^* \Leftrightarrow z \in \mathbb{R}$
(b) $z + z^* = 2 \operatorname{Re}(z)$	(f) $(z + w)^* = z^* + w^*$
(c) $z - z^* = 2i \operatorname{Im}(z)$	(g) $(zw)^* = z^* w^*$
(d) $zz^* = x^2 + y^2$	

Finding $\arg(z)$:

Consider a point P on an Argand diagram representing $z = x + iy$, $x, y \in \mathbb{R}$

Always first identify the quadrant that z lies, then let **basic angle**, $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$.

1 st quadrant	2 nd quadrant	3 rd quadrant	4 th quadrant
 $\arg(z) = \alpha$	 $\arg(z) = \pi - \alpha$	 $\arg(z) = -(\pi - \alpha)$	 $\arg(z) = -\alpha$

Note:

- For **REAL** number x ,
 - (a) $\arg(x) = 0$ if $x > 0$
 - (b) $\arg(x) = \pi$ if $x < 0$
- For **PURELY imaginary number** $z = iy$,
 - (a) $\arg(z) = \frac{\pi}{2}$ if $y > 0$
 - (b) $\arg(z) = -\frac{\pi}{2}$ if $y < 0$

Geometrical Effect of Multiplying a Complex Number z by i

- It is an anti-clockwise rotation of z through an angle of $\frac{\pi}{2}$ radians about origin O .

Geometrical Effect of Conjugation of a Complex Number z

- It is a reflection of the complex number z about the real axis.

Geometrical Effect of Negation of a Complex Number z

- It is a rotation of the complex number z through an angle of π radians about origin O .

Complex Roots of Polynomials

Reminder: a polynomial of degree n has n roots (complex and/or real).

If a polynomial with **REAL coefficients** has non-real roots, they occur in conjugate pairs.

(Note: This result **does not apply** if at least one of the coefficients is a non-real number.)

Application:

$-3 - 2i$ is a **root** of the equation $P(z) = z^3 + az^2 + bz - 13$, where $a, b \in \mathbb{R}$

$\Rightarrow -3 + 2i$ is also a **root** of the equation $P(z) = 0$.

Then $[z - (-3 - 2i)]$ and $[z - (-3 + 2i)]$ are both **factors** of the polynomial $P(z)$.

Using the above two **linear factors**, we can form the following **quadratic factor**:

$$\begin{aligned} & \text{Quadratic factor} \\ &= [z - (-3 - 2i)][z - (-3 + 2i)] \\ &= (z + 3 + 2i)(z + 3 - 2i) \\ &= (z + 3)^2 - (2i)^2 \\ &= z^2 + 6z + 13 \end{aligned}$$

SOME COMMON TYPES OF PROBLEMS

1	Solving an equation with complex number as coefficient(s) or simultaneous equations involving complex numbers, by comparing real and imaginary parts respectively.
2	Solving an equation with real coefficient(s): one may start with z and its conjugate z^* , to find roots and/or to find unknown coefficients.

Revision Tutorial Questions

- 1 Verify that $z = -i$ is a root of the cubic equation $z^3 + i(1 - \sqrt{3})z^2 + (\sqrt{3} - 1)z - i = 0$. Hence, find the other roots of the equation in exact form, and represent all the roots on an Argand diagram.

[5]

Source of Questions: NYJC Prelim 9758/2018/02/Q3(i)

- 2 Do not use a calculator in answering this question.

The equation $z^3 + 4z^2 + 8z + 8 = 0$ has a root $z = -2$. Find the other two roots z_1 and z_2 .

[2]

Source of Question: YJC Prelim 9758/2018/01/Q3(i)

- 3 The complex number z is given by $z = 1 + ia$, where a is a positive real number.

Given that the argument of $(z - 3)$ is $\frac{3\pi}{4}$, find the value of a .

[2]

Source of Question: MJC Prelim 9758/2018/02/Q1

- 4 (a) Consider the polynomial $P(z) = a_0 + a_1z + a_2z^2$, where $a_0, a_1, a_2 \in \mathbb{R}$, $a_2 \neq 0$. Show that if the equation $P(z) = 0$ has a complex root w , then its complex conjugate w^* must also be a root to the equation.

[3]

- (b) The polynomial $P(z)$ has real coefficients. The equation $P(z) = 0$ has a root $z_1 = -a + ai$, where a is a positive real number.

(i) Write down a second root, z_2 , in terms of a .

[1]

(ii) Hence sketch the points Z_1 and Z_2 , representing z_1 and z_2 respectively, on an Argand diagram.

[2]

(iii) Deduce the area of OZ_1Z_2 , in terms of a .

[1]

Source of Questions: DHS Prelim 9758/2018/01/Q7(b)

5 When the polynomial $ax^4 + bx^3 + cx^2 + 24x - 44$, where a, b and $c \in \mathbb{R}$, is divided by $(x-1)$, $(x+1)$ and $(x-2)$, the remainders are -18 , -54 and 0 respectively.

(i) Find the values of a, b and c . [3]

(ii) The equation $ax^4 + bx^3 + cx^2 + 24x - 44 = 0$, with the values of a, b and c found in part (i), has a root $3 - (\sqrt{2})i$. Find the other roots of the equation, showing your working clearly. [3]

Source of Questions: SAJC Prelim 9758/2018/01/Q5

6 Do not use a graphic calculator in answering this question.

(i) By letting $z = a + bi$, solve $z^2 = i$, giving your answers in exact form. [3]

(ii) Solve the equation $w^2 + 2w + (1 - 8i) = 0$, giving the roots in Cartesian form. [3]

(iii) Hence, solve $(1 - 8i)v^2 + 2iv - 1 = 0$, giving your answers in Cartesian form. [3]

Source of Questions: MI Promo 9758/2018/PU2/01/Q5

7 The complex numbers z and w satisfy the following equations

$$z - wz^* = 6 + 22i$$

$$2w + 3z = 20$$

(i) Find z and w in the form $a + bi$, where a and b are real, $a \neq 0$. [5]

Let P and Q be the points representing z and w respectively.

(ii) Label the points P and Q on a single Argand diagram. Deduce the angle POQ . [2]

Source of Question: RVHS Prelim 9758/2018/01/Q8(b)

8 Given that $f(z) = z^4 + 2z^3 + (\pi^2 + e^2)z^2 + pz + q$, where $p \geq 0$ and $q \geq 0$, and that $i\pi$ satisfies the equation $f(z) = 0$.

(i) Find the exact values of p and q . [2]

(ii) Hence find the remaining roots of $f(z) = 0$ in the form $\alpha + \beta i$, where α and β are exact real numbers. [4]

Source of Questions: CJC Prelim 9758/2018/01/Q11(modified)

- 9 It is given that $z = -\frac{1}{2}$ is a root of the equation

$$8z^3 + (4 - 4\sqrt{2})z^2 + (2 - 2\sqrt{2})z + 1 = 0.$$

The roots of the equation are denoted by z_1, z_2, z_3 , where $\arg(z_1) < \arg(z_2) < \arg(z_3)$.

- (i) Find z_1, z_2 and z_3 in the form $a + bi$. Write down the modulus and argument of z_2 .
[7]

Given the complex number z_4 has modulus 4 and argument $\frac{\pi}{3}$.

- (ii) Mark, on an Argand diagram, the points Z_2, Z_3 and Z_4 . [2]
(iii) By considering $\sin(A - B)$ with suitable values of A and B , show that

$$\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{2}}{4}(\sqrt{3} - 1). \quad [1]$$

Hence or otherwise, find the exact area of the quadrilateral $OZ_3Z_4Z_2$. [2]